Part 4 Brief · Evals

mcp-ops-server · 18 Oct 2026 · PR #5 · Status: 31/31 tasks pass, 7/7 negative controls caught, awaiting Codex review

WHAT WAS BUILT

- **31 golden tasks** in 7 categories (lookups, bookings, referral bookings, refusals, cancellations, auth, conflicts). Each has a prompt, who it runs as (staff, read-only, or a patient-bound portal token) and what must and must not happen.
- **A trajectory grader** that checks every step, not just the final answer: required steps in order, forbidden steps, no touching other patients, call and error budgets, and the **exact change in the database**.
- **A replay agent** (no LLM, free, deterministic) that runs in CI, plus `evals/report.md`: per-task results, pass rate by category, tool latency.
- **7 negative controls:** deliberately bad runs (wrong patient, skipped referral check, writes during a lookup, flailing) that the grader must fail. Together they cover every check.

HOW IT WORKS

Piece	Role
harness.py	Starts the real server with issued tokens and a file database; agents talk to it over MCP like any client. Diffs the appointments table before and after each task.
grader.py	Six checks per task. final_state compares the actual database change to the expected bookings and cancellations, so a correct-sounding but wrong run still fails.
agents.py	One Agent interface. Replay now; the Part 5 LLM agent plugs into the same slot and is graded identically.

WHY IT WAS DONE THIS WAY

- **Grade the database, not the reply.** An agent can say "booked!" and book the wrong patient. The diff catches that; a final-answer check would not.
- **Grade the steps.** Booking with a valid referral ID but never checking its status can still succeed. Only step-level grading sees that the check was skipped.
- **Evals use real auth.** 7 of the 64 calls are expected refusals (scope, patient binding, double booking), so the suite also tests Part 3's rules end to end.

WHAT WENT WRONG OR SURPRISED ME (LOGGED IN FAILURES.MD)

- **A loosened rule was invisible to every golden task.** I made pending referrals bookable as a test; all 31 tasks still passed, because a well-behaved agent never tries it. Only a negative control failed. Now the controls must cover every check.
- **Controls crashed on a shared database** by reusing a slot the golden run had booked. They now get a fresh server, and a broken setup fails one task instead of the run.

VERIFY IT YOURSELF

```
python -m evals.run       # 31/31 tasks passed; 7/7 negative controls caught
cat evals/report.md       # per-task results, categories, latency
ruff check . && python -m pytest -q   # 150 passed (includes the full suite)
```

INTERVIEW TALKING POINT

"My eval grades what the agent did, step by step, and what changed in the database, not what it said. And I test the grader: deliberately bad runs must fail. One caught a server regression that every well-behaved task missed."

NEXT: PART 5, CI GATE AND REAL NUMBERS

A dedicated eval job that blocks merges, and a live LLM agent on the same 31 tasks to measure pass rate, cost and latency. That needs an API key as a secret, your call. **Your action:** paste the Codex prompt, then merge PR #5 once fixes are in.